

Telemedicine At A Glance

Telemedicine can be broadly defined as healthcare services delivered through telecommunication networks. In simplest terms, telemedicine can be explained as two doctors or doctor-patient discussing the case on a phone. Practiced telemedicine is more than simple voice communication over telephone lines. It is the integration of store-and-forward technology with real-time responses.

Telemedicine is the only solution to provide healthcare services to people in remote locations. Conventionally, the government would need to build hospitals in every village and provide physicians to each, which is an extremely daunting task. Alternatively, we could use IT tools like telemedicine to provide quality healthcare to all.

Technical components of a typical telemedicine system

1. Data acquisition (the capability to integrate with medical equipment and acquire data)
2. Electronic record management (includes patient registration details, history, prescription, lab reports, etc)
3. Video and image conferencing
4. Communication networks (broadband, mobile, etc) for data transfer between remote locations and expert centres

Non-technical components

1. The patient
2. A doctor situated at the remote location who decides the need for telemedicine and links up with the expert for consultation through the telemedicine system
3. A technical operator who handles the telemedicine system, voice or video-based conferencing, report transmission, prescription receipt, print-outs, etc
4. An expert who caters to the telemedicine request from a main centre

Benefits of telemedicine

1. Healthcare services delivered at your doorstep. In urban areas, this means avoiding the traffic. In rural areas, it means the boon of healthcare. The same can be true for second opinions, lab reports, radiology reports, etc.
2. Telemedicine components like Web-based patient health records can be very useful in the case of emergencies.
3. Patient health records prove to be a lifetime medical history and can be used for health monitoring. Prompt reminders for health checkups may prevent many issues.
4. In the case of chronic cases, repeated hospital visits can be prevented.
5. Improved reach to rural areas.
6. Can even be used for healthcare education.

—By *Devendra Patel, director, Medisoft Telemedicine*

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After painstaking research in labs and on the field, ReMeDi was born. This remote medical diagnostic solution enabled doctors to videoconferencing with patients who came to seek their services through Internet centres set up in the villages. Innovative technologies developed by TeNeT helped to bridge the last-mile gap. Doctors could remotely view the output of probe-fitted medical equipment like electro-cardiograms (ECG), blood pressure monitors and stethoscopes administered by the local operators, store and analyse medical records online, as well

as prescribe medicines. In all, it was almost as if the patient and doctor were sitting across a desk.

It was not all about technology though—the duo had to build a viable business model that split up the healthcare delivery model into various levels ranging from the doctor to the local Internet centre operator to the pharmacists and diagnostic labs, and ensure that it was profitable for them all to stay in business. Plus, they had to change the firm cultural mindsets of people to make them accept a local operator as a healthcare practitioner. ReMeDi is now being used in states like Uttar Pradesh, Bihar and Tamilnadu, and also being

piloted in other countries.

Madurai-based Aravind Eye Care is a highly-reputed institution in India. It is known for its research as well as expert panel of doctors who have treated several seemingly-hopeless cases. More than anything else, they are known for their reach at the bottom of the pyramid, all over the country. While they used to cover this population through eye camps in the past, now they have turned to telemedicine as a more permanent solution. Dr S. Aravind, director of Aravind Eye Care, reasoned in a recent interview, "We realised that camps happen when we choose them to happen, not when the disease happens."

They then set up several vision centres to reach out to a population of 50,000. Since the localised communities could not afford to appoint ophthalmologists, Aravind posted their own technicians to handle the centres. This gave them the idea of connecting the centres with the base using telemedicine technology from the University of California, Berkeley. They now have about 32 centres connected on a hub-and-spoke model to the base using broadband services available in the areas that they operate in. They have their own telemedicine software, for example, Adres for diabetic retinopathy and EyesTalk to maintain medical records. With these and other solutions, Aravind's doctors sitting in a clinic in the town of Theni are able to treat a villager in Aandipatti, a tiny dot in the map, who seeks consultancy through a wireless videoconferencing facility.

Last month, the Nanavati Hospital in Mumbai adopted eUNO R10—a mobile ECG application launched by Maestros Mediline Systems, a maker of high-tech medical equipment, in association with Vodafone. eUNO R10 enables cardiologists to monitor patients' ECG reports on their BlackBerry smartphones. The instrument helps to diagnose and treat heart attacks almost immediately. Without such a mobile diagnosis, the symptoms of a heart attack could easily be mistaken for that of